Summary in Graph

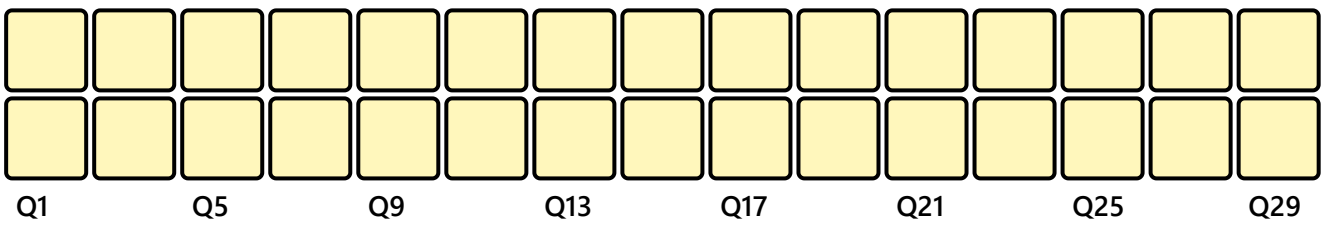
Exam Summary (GATE Overflow CS Test Series 2026 | Data Structures | Shallow Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



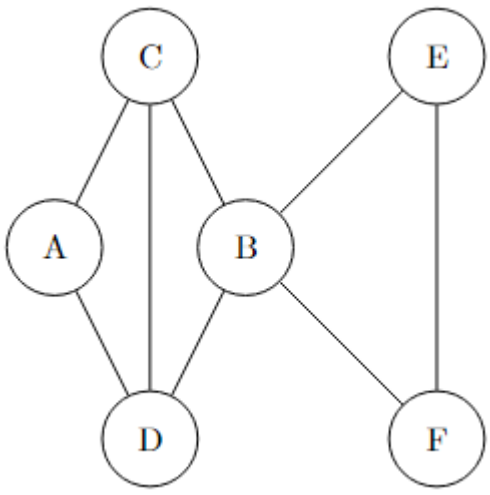
* Average Stats based on first-attempt stats of Top 10% (2) out of 24 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

The Cut Vertices in the given graph are:



A. D, B and C
B. D and C
C. B
D. A, C and E

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Sarah is tasked with writing a program to construct a binary tree from the given preorder and inorder traversals, where the tree has n nodes. In this case, she is not allowed to use any additional data structures like hashmaps or dictionaries. What is the best possible time complexity for this program, considering the constraints?

A. $\Theta(n \log n)$
B. $\Theta(n)$
C. $\Theta(n^2)$
D. $\Theta(n^3)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Evaluate the result of the postfix expression 5 3 2 ^ 4 * + 6 2 / -

Your Answer: | Correct Answer: 38:38 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Which data structures are most appropriate for performing preorder traversal and reverse level order traversal of a binary tree, respectively?

A. (Stack) and (Queue + Stack)

B. (Queue) and (Stack + Queue)

C. (Stack) and (Stack + Stack)

D. (Queue) and (Queue + Queue)

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following infix expression:
$A + (B \times C) - \frac{D}{E} \times (F + G)$
Which of the following represents the correct prefix form of the expression?

A. $- + A \times BC \div DE + FG$

B. $- + A \times BC \times \div DE + FG$

C. $+ - A \times BC \div DE \times FG$

D. $+ - A \times BC \div \div DEFG$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

In a binary search tree, you are tasked with searching for the key 57. Which of the following traversal sequences represent the valid order of nodes visited while searching for 57?

A. $45, 78, 34, 56, 57$

B. $45, 56, 34, 57$

C. $78, 45, 34, 56, 57$

D. $78, 34, 45, 56, 57$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a directed graph G with v vertices and e edges. Which of the following statements is true about the number of strongly connected components (SCCs) in the graph?
A. The number of strongly connected components is always less than or equal to v.
B. The number of strongly connected components is always equal to $e - v + 1$.
C The number of strongly connected components is always equal to v.
D. The number of strongly connected components is always equal to e.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider a set of elements $X = \{P, Q, R, S, T\}$ that are inserted into both a stack and a input restricted deque in order. One element at a time is removed from each structure independently. Deletions can occur at any point after third insertion is done and insertions in deque happens only at the rear end while, deletions in

input restricted deque happens at both ends.
Which of the following sequences is possible for the order in which the elements are removed from the stack and the queue, respectively?

A. $RSTQP, RTSQP$
B. $RQPST, TSRQP$
C. $QPRST, PTSRQ$
D. Both A and B

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Given the following array-based representation of a binary tree with $n = 10$:
$[10, 20, 30, 40, 50, 60, 70, 80, 90, 100]$, where the root is indexed as 1.Find the index and value of the right child of node $i = 4$, if it exists.

A. $10, 100$
B. $7, 90$
C. $8, 20$
D. $9, 90$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Which of the following statements is TRUE regarding XOR linked list?

A. Memory requirements of doubly-linked list is not greater than XOR linked list.
B. XOR-linked lists does not have the advantages of both singly and doubly linked lists.
C. In subtraction linked list, the entire list can be relocated in memory without needing any patching of pointer values.
D. A memory efficient version of Doubly Linked List cannot be created using only one space for address field with every node.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose you have a circular doubly linked list C containing r elements and another singly linked list P containing m nodes, where each node in P stores a direct reference to a node in C (with $r/3 < m < r/2$). The references in P are not in any specific order. Determine the most efficient algorithm, using $O(1)$ additional space if necessary and assuming only comparison-based sorting, to output the elements from C as specified by the references in P. The time complexity of the best algorithm will be:

A. $\Theta(r)$
B. $\Theta(m \log m)$
C. $\Theta(r \log m)$
D. $\Theta(r + m)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

The time complexity of below code is

```
#include <stdio.h>
#define TEN 10
void updateFreq(int n, int fq[])
{
    while (n)
    {
        int digit = n % TEN;
        fq[digit]++;
        n /= TEN;
    }
}
int areAnagrams(int a, int b)
{
    int fqA[TEN] = {0};
    int fqB[TEN] = {0};
    updateFreq(a, fqA);
    updateFreq(b, fqB);
    for (int i = 0; i < TEN; i++)
    {
        if (fqA[i] != fqB[i])
        {
            return 0;
        }
    }
    return 1;
}
int main()
{
    int a = 240, b = 204;
    if (areAnagrams(a, b))
    {
        printf("Yes\n");
    }
    else
    {
        printf("No\n");
    }
    return 0;
}
```

A. $O(\log_{10} a + \log_{10} b)$
B. $O(a + b)$
C. $O(b \log_{10} a)$
D. $O(b)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

```
void cD(struct node* start)
{
    if (start == NULL || start->next == NULL)
        return;
    int dif = start->next->data - start->data;

    printf("%d\n", dif);
    cD(start->next);
}
```

$7 \rightarrow 14 \rightarrow 21 \rightarrow 28 \rightarrow 35 \rightarrow 42$

Using the above function, what is the sum of the differences printed when the function is called with the head of the above list?

Your Answer: | Correct Answer: 35:35 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider the following C function to delete a node at the end of linked list.

```c
void delete_node_end(struct node *Start)
{

    struct node *p;

    if(Start == NULL)
    {
      printf("Llist empty");
      return;
    }
    while (Start->next != NULL)
    {
      p=Start;
      Start = Start->next;
    }
    p->next = NULL;
    free(Start);
    printf("%d", Start->data);
}
```

Which of the following is $TRUE$ about above function?

A. The function may crash when the passed linked list is empty.
B. The function does not delete a node at the end of linked list.
C. Pointer Start becomes dangling pointer when a node is deleted at the end of linked list.
D. Last printf prints the data of initial node of the linked list.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Two binary trees having p and q elements respectively, whereas $p >= 1$ and $q >= 1$. Both of these trees are merged into another binary tree in $O((n)^a(\log(n))^b)$ time where n in this case is the total number of nodes that will be present in the new merged tree, Find the value of $5 * a - 2 * b =$ ____

Your Answer: | Correct Answer: 5:5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

You are given the following code snippet for inserting a new element into a max-heap, where the parent(i) function returns the index of the parent node, and the size variable holds the current number of elements in the heap.

```
void swap(int *p, int *q)
{
 int temp = *p;
  *p = *q;
  *q = temp;
}
void insert(int element)
{
    size++;
    int i = size - 1;
    heap[i] = element;

    while (_________________)
        {
        swap(&heap[i], &heap[parent(i)]);
        i = parent(i);
        }
}
```

The correct replacement for While condition will be?

A. $i > 0$ && $heap[parent(i)] < heap[i]$

B. $i < 0$ || $heap[parent(i)] > heap[i]$

C. $i < 0$ && $heap[parent(i)] > heap[i]$

D. $i > 0$ || $heap[parent(i)] < heap[i]$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Given an array of elements $6, 12, 15, 9, 20, 30, 10, 18$, which of the following is not a valid array implementation of max-heap of these elements?

A. $30, 20, 15, 9, 12, 6, 10, 18$
B. $20, 18, 15, 9, 12, 6, 10, 30$
C. $30, 20, 10, 9, 12, 6, 15, 18$
D. All of the above are not Max-heaps

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Let $f(n), g(n)$, and $h(n)$ represent the time complexities of three different algorithms. Suppose $f(n) = \Theta(g(n))$ and $h(n) = \Omega(g(n))$. Which of the following is necessarily FALSE?

A. $f(n) = O(h(n))$
B. $g(n) = \Theta(f(n))$
C. $h(n) = O(f(n))$
D. None of the above are necessarily FALSE

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

What is the maximum number of nodes in a complete k-ary tree where $k = 3$ (each node has at most 3 children) and the height (starting from 1) is 5? The maximum number of nodes is _____.

Your Answer: | Correct Answer: 121:121 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

What is the best time complexity (not considering best-case input but the most efficient algorithm for all cases) for finding the kth smallest element in a binary search tree with n nodes and height h?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $O(h)$
D. $\Omega(n)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

In an AVL tree with n nodes, if the time complexity for performing an insertion operation, including all necessary rotations to maintain the balance of the tree, expressed as $\Theta(n^p(\log n)^q)$ based on the size and height of the tree, then what is the value of $3^p + 4^q$?

Your Answer: | Correct Answer: 5:5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 Numerical Type | Award: 2 | Penalty: 0 | Data Structures

In a binary search tree, the following key values (not necessarily in the order given) are encountered while searching for the key 42:

$$5, 12, 18, 25, 30, 38, 60, 75.$$

What is the total number of distinct binary search trees that can be constructed using these keys, given that the keys are visited in some order during the search for 42?

Your Answer: | Correct Answer: 264:264 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

What will be the output of the following C program?

```c
#include<stdio.h>
int main()
{
    int a[2][3] = {{1, 2, 3}, {4, 5, 6}};
    int (*ptr1)[3] = a;
    int (*ptr2)[3] = a + 1;
    printf("%d %d ", *(*ptr1 + 1), *(*(ptr1 + 1) + 1));
    printf("%d %d ", *(*ptr2 + 2), *(*(ptr2 - 1) + 2));
    printf("%d %d\n", *ptr1[0], *ptr2[0]);
    return 0;
}
```

A. 2 4 6 3 1 4

B. 2 5 6 3 1 4

C. 2 3 6 5 1 2

D. 2 5 6 1 4 2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24

Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following representation of a binary tree where A: B, C denotes that B and C are the left and right child nodes of A respectively.

$10 : 20, 30$
$20 : 40, 50$
$30 : 60, 70$
$40 : NULL, NULL$
$50 : NULL, NULL$
$60 : NULL, NULL$
$70 : NULL, NULL$

If the root of the above tree is passed to the following modified C function (assume that the left child is child1 and the right child is child2 and all NULL values are assigned properly), the return value will be ____

```c
int foo(struct node *root)
{
    if(!root) return 0;
    int left = foo(root->child1);
    int right = foo(root->child2);
    return left + right + 1;
}
```

Your Answer: | Correct Answer: 7:7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Given three algorithms with time complexities $p(n)$, $q(n)$, and $r(n)$, where:

- $p(n)$ grows slower than $q(n)$ as $n \to \infty$,
- $r(n)$ grows faster than $q(n)$ as $n \to \infty$,

Which of the following is FALSE?

A. $p(n) = O(q(n))$
B. $r(n) = O(q(n))$
C. $p(n) = o(q(n))$
D. $r(n) = \Omega(q(n))$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Given an array:

$$4,\ 1,\ 3,\ 10,\ 2,\ 7,\ 9,\ 5$$

What is the least number of swaps required to rearrange the array into a max-heap, where each parent node is greater than or equal to its child nodes?

Your Answer: | Correct Answer: 5:5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Determine the Correct Asymptotic Growth Order; Given the following functions, determine the correct order of their asymptotic growth rates as $n \to \infty$:
$\log n$, $n \log n$, 2^n, $n^{\log n}$, $\log(\log n)$, n^{2^n}, $\sqrt{n}$
Which of the following correctly arranges these functions from slowest to fastest growth?

A. $\log(\log n) < \log n < \sqrt{n} < n \log n < n^{\log n} < 2^n < n^{2^n}$
B. $\log n < \log(\log n) < \sqrt{n} < 2^n < n \log n < n^{\log n} < n^{2^n}$
C. $\log(\log n) < \sqrt{n} < \log n < n \log n < 2^n < n^{\log n} < n^{2^n}$
D. $\log n < \log(\log n) < \sqrt{n} < n^{\log n} < n \log n < 2^n < n^{2^n}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

The best data structures in terms of asymptotic time complexities for performing the following operations respectively are:
1. UNION(A, B): Combine all unique elements from sets A and B.
2. Insert(T, key): Insert a key into data structure T.
3. Find Min(T): Retrieve the minimum value from T.
4. Range Query(T, x, y)(considering it as a primary operation for T): Find all elements between x and y in T.
Choose the correct sequence of data structures for optimal performance:

A. Hash Table, Binary Heap, Binary Search Tree, AVL Tree
B. Disjoint Set, Binary Heap, Sorted Array, Binary Search Tree
C. Disjoint Set, Hash Table, Binary Heap, Segment Tree
D. Hash Table, AVL Tree, Binary Heap, Sorted Array

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a QUEUE implemented using two STACKS, where ENQUEUE operations favor pushing onto one stack and DEQUEUE operations involve transferring elements between the stacks. Given that k ENQUEUE operations and l DEQUEUE operations are performed on this queue, where k and l are much smaller than the total number of elements n initially in the queue, what is the amortized time complexity for performing these operations?

A. $O(k+l)$
B. $O(k \cdot l)$
C. $O(k+l+n)$
D. $O(n)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following three structure declarations and their initialization in C language:

```c
struct point1
{
    int x, y, z;
} mypoint1[3000];

for(i = 0; i < 3000; i++)
{
    mypoint1[i].x = rand();
    mypoint1[i].y = rand();
    mypoint1[i].z = rand();
}

struct point2
{
    int x[3000], y[3000], z[3000];
} mypoint2;

for(i = 0; i < 3000; i++)
{
    mypoint2.x[i] = rand();
    mypoint2.y[i] = rand();
    mypoint2.z[i] = rand();
}

struct point3
{
    int x[3000];
} mypoint3;

for(i = 0; i < 3000; i++)
{
    mypoint3.x[i] = rand();
}
```

Assume the following:

- The size of an integer is 4 bytes,
- Cache line size is 128 bytes,
- Cache is 64 KB, direct-mapped, with all cache lines invalidated initially,
- All arrays and structures are aligned to cache lines, with no other intervening cache accesses.

Let n_1, n_2, and n_3 represent the number of cache misses during the initialization of $mypoint1$, $mypoint2$, and $mypoint3$, respectively. Compute the sum $n_1 + n_2 + n_3$.

Your Answer: | Correct Answer: 676:676 | Not Attempted | Time taken: 00min 00sec | Discuss

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